

EDUCATION

Tulane University, School of Science and Engineering
B.S. in Mathematics, GPA: 3.97 / 4.00

New Orleans, LA
August 2024 – May 2027, Expected

RESEARCH INTERESTS

Post-quantum and code-based cryptography; algebraic coding theory; quantum information theory; algebraic geometry in cryptography.

PUBLICATIONS & PREPRINTS

- [1] M. B. Can and E. C. Naig, “A Theory of Leakage in Secret Sharing and Evaluation Codes”, Submitted, under review, 2026.
- [2] M. B. Can and E. C. Naig, *Applications of Bruhat-Chevalley-Renner Decomposition to Metric-Aware Code-Based Cryptography*, Cryptology ePrint Archive, Paper 2026/665, Preprint, 2026. [Online]. Available: <https://eprint.iacr.org/2026/665>
- [3] M. B. Can and E. C. Naig, “Module-HQC”, Submitted to IEEE Transactions on Information Theory, 2026.
- [4] E. C. Naig and S. A. Fenner, *Parallel Repetition in the Two-Player Quantum Cloning Game*, arXiv:2608.00930 [quant-ph], Preprint, 2026. [Online]. Available: <https://arxiv.org/abs/2608.00930>

TALKS & PRESENTATIONS

- *Parallel Repetition in the Two-Player Quantum Cloning Game*. Talk presented by co-author S. A. Fenner, Ulm University, Ulm, Germany ([event page](#)). September 2026
- *Quantum Position Verification and the Quantum Cloning Game*. Poster, USC Summer Research Symposium, University of South Carolina. July 2026
- *Quantum Position Verification: Cloning Games and an Open Repetition Problem*. Talk to faculty and students of the NSF REU in Quantum Information Science, University of South Carolina. July 2026

PATENTS

- **Tunable Module Hamming Quasi-Cyclic Code-Based Cryptography** Filed August 28, 2026
U.S. provisional patent application no. 64/143,287; co-inventors: Mahir Bilen Can and Eli Coe Naig. Attorney docket no. 752001-8030.
- **Systems and Methods for Secure and Reliable Data Storage and Transmission** Filed April 2026
PCT patent application; co-inventors: Mahir Bilen Can, Dillon Montero, and Eli Coe Naig. Tulane Ref. No. 2025-010-02; Foley Ref. No. 136669-0184.

- **Metric-Aware Code-Based Cryptography**

Filed March 6, 2026

U.S. provisional patent application; co-inventors: Mahir Bilen Can and Eli Coe Naig. Attorney docket no. 752001-8010.

RESEARCH EXPERIENCE

Undergraduate Researcher

New Orleans, LA

Advisor: Mahir Bilen Can, Tulane University

August 2025 – Present

- Designed code-based public-key cryptosystems, including a quasi-circulant trapdoor for Niederreiter-type key encapsulation.
- Extended code-based cryptography beyond the Hamming metric to the NRT and sum-rank metrics, using code isometry groups and Bruhat-Chevalley-Renner decompositions.
- Proved an NP-hardness theorem and gave decoding-hardness reductions underpinning the schemes' security.
- Developed a matroid-based theory of information leakage in linear secret sharing and evaluation codes.

NSF REU in Quantum Information Science

Columbia, SC

Advisor: Stephen Fenner, University of South Carolina

May 2026 – August 2026

- Studied quantum position verification at USC's NSF REU site (NSF award 2447817).
- Disproved strong parallel repetition for the quantum cloning game and improved its best known upper bound (see Publications).

RTG Summer Math Program

Baton Rouge, LA

Louisiana State University

May 2025 – June 2025

- Studied hyperplane arrangements and matroid theory in the workshop funded by LSU's NSF Research Training Group grant (NSF award DMS-2231492).
- Prepared and delivered lectures on hyperplane arrangements, geometric lattices, and characteristic polynomials.
- Led peer discussions linking combinatorial geometry to coding theory.

TECHNICAL & ENTREPRENEURIAL EXPERIENCE

QSPARC Labs

New Orleans, LA

Co-Founder, CEO, and Lead Developer

August 2025 – Present

- Co-founded a startup developing post-quantum cryptography grounded in original coding-theory research.
- Build reference implementations of the published schemes in Python, Rust, and C, with test suites and reproducible benchmarks.
- Verified a post-quantum key-establishment protocol with machine-checked symbolic security proofs in ProVerif.
- Direct the company's patent work: three applications filed, with more in preparation (see Patents).
- Lead and mentor a team of four: two engineering interns and two part-time developers.
- Accepted into an NSF-funded I-Corps-style commercialization program administered through MIT.

TEACHING & MENTORING

- **Volunteer Tutor, Excite All Stars**

Fall 2026 – Present

Tutor middle school students in mathematics and computer science for three hours each week at Lake Forest Charter School in New Orleans East, through Excite All Stars in partnership with Tulane's service-learning program.

- Volunteer Tutor, Tulane University Fall 2025 – Present
Tutor student-athletes in linear algebra and abstract algebra, supporting coursework alongside practice and competition schedules.

AWARDS & GRANTS

- Tulane Honors Scholar 2024–Present
- Dean’s List, Tulane University Fall 2024; Spring 2025; Fall 2025; Spring 2026
- CELT Faculty-Mentored Undergraduate Research Grant, Tulane University; mentor: Mahir Bilen Can. 2026
- Offered a place in the Newcomb-Tulane College Summer Research Program; declined in favor of the NSF REU. 2026
- Chosen for a Tulane Innovation Institute program on research translation and proposal development. 2025
- Ellen Applebaum Temkin Scholarship, Mathematics Department, West High School 2024

LEADERSHIP & SERVICE

- President, Tulane Math Club 2026–2027
Lead programming and organization for the university math club.
- Co-Founder and Vice President, Tulane Science Olympiad 2024–Present
Co-founded Tulane’s first Science Olympiad invitational; coordinated logistics for an event serving 380 students from nearly 20 schools and developed tulanescioly.org.
- Trip Leader, Tulane Outdoor Adventures 2024–Present
Lead outdoor trips for student groups, managing route planning, risk assessment, logistics, and participant safety.

TECHNICAL SKILLS

- **Programming:** Python, Rust, C, C++, Java, JavaScript, TypeScript, MATLAB, R
- **Tools:** Git, GitHub, L^AT_EX, Linux, ProVerif
- **Languages:** English (native), Chinese (intermediate reading/writing; beginner speaking)

SELECTED ADVANCED COURSEWORK & INDEPENDENT STUDY

- **Selected Advanced Coursework:** Abstract Algebra I–II, Real Analysis I–II, Linear Algebra, Probability, and Combinatorics.
- **Planned Fall 2026 Coursework:** Algebra I, Algebraic Geometry, Mathematical Foundations of Computer Security, and Honors Thesis.
- **Independent Study:** Algebraic geometry, error-correcting codes, real analysis, and manifolds; texts include Hartshorne, Huffman–Pless, Rudin, and Spivak.